Clone coverage
near-clone families
(`family_id`)

GitSkills tables (Parquet)

artifacts
`file_sha`
`+ inclusion fields (Fig. 2)`
`repo_full_name, path`
`location_class`
`content`
`history_fetched`
`commit_count`
`first_commit_at`
`last_commit_at`
`first/last_commit_author`
`first/last_commit_author_type`
`first/last_commit_message`

repos
`full_name`
`created_at`
`pushed_at`

mining_runs
`finished_at`

Pre-processing

- Keep one row per distinct skill (same rule as Figure 2)
- Parse commit times as UTC
- Flag AI co-authored commits and bot authors
- Sample edited skills (`commit_count` > 1), stratified by `location_class`

Cross-copy churn

- Order each family's members by `first_commit_at`, else `repos.created_at`
- Parent = most similar earlier member (token similarity)
- Diff each member with its parent: tokens added and removed
- Links below 0.5 similarity are dropped; near-copy ≥ 0.9, variant 0.5–0.9

Activity churn

- Edits = `commit_count` − 1, and edits per month
- Active span, and dormancy up to the snapshot (`finished_at`)
- Author change, AI co-authorship, and dormancy in an active repo (`pushed_at`)

Token churn

- Sampled skills only: follow the file's history through renames (`git log --follow`)
- Tokenize every version with the Anthropic tokenizer
- Diff consecutive versions: tokens added and removed, relative to the earlier version's size

Lineage links

per link
`parent_sha, child_sha`
`similarity, band`
`tokens_added`
`tokens_removed`
`direction_basis`

Churn columns

per file_sha
`edits`
`edits_per_month`
`active_span_days`
`dormancy_days`
`author_changed`
`ai_coauthored`
`tokens_added`
`tokens_removed`
`relative_churn`

GitHub
git clone of the sampled repositories